

CASINO BUSINESS PRESENTATION

SPIT SHINE

The 5-Seat Table Game of Show, Spit, Shine & Win

Complete Deal & Play Procedures, Rules, Odds, Audit Data, and Integrity Controls



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1. Executive Summary

Spit Shine is a proprietary 5-seat casino table game played with a 104-card shoe (two standard 52-card decks). Each seat receives four cards and, across seven clearly defined phases - Private Reveal, Showing, Bow Out, Spit, Shine, Selection, and Showdown - builds the highest-scoring two-card hand. The game combines the bluffing depth of poker, the fixed-dealer structure of blackjack, and the clean point-comparison of baccarat, but its signature mechanics (Spit card exchange and Shine redraw) create a level of player agency absent from most table games.

This guide is the complete operating manual for the game: how to deal it, how to play it, every rule and resolution, the full statistical audit from every perspective (player, dealer, house, regulator), and the integrity controls required for a regulated casino environment. Every figure in this document is reproducible from the canonical engine and the audited simulation dataset.

Key commercial metrics

Metric	Value	Note
Game type	5-seat point-comparison table game	1 human-capable seat + 3 AI/player seats + dealer seat
Equipment	2 standard 52-card decks (104 cards)	Fresh shuffled shoe each round; cut card recommended
Ante	100 chips (table minimum)	House limits configurable
House take	25% of pot (rake)	Bow Out refunds reduce the pot before rake
House edge (competitive)	24.5%	Measured over 20,000 rounds; ~25% per ante by rule
Dealer win rate	24.6%	Fair baseline for 5 seats is 20%
Player win rate (seat 0)	17.8%	Seat-position effect documented in Section 13
P(Double Aces in 4 cards)	2.90%	Exact; A+A scores 1000 points
P(someone dealt Double Aces)	13.92%	Per 5-seat round, exact Monte Carlo
Exact-tie rounds	3.2%	Pot splits among tied winners
Rounds per hour (model)	17-30 depending on pace	Section 11 - validate in floor trials
Illustrative daily table win	\$18.7k-\$66.2k per table (24.5% hold, \$1 chips)	Section 11 - model assumptions apply
Cooperative player EV	-46.1 / round	Section 13.5 - needs rebalancing for casino floor

The audit's headline finding is that the game's published odds were materially incorrect (the Double-Aces frequency was understated by roughly 30x and the house edge by 5x). This guide presents the corrected, simulation-verified numbers and the canonical rule set that every future implementation must follow.

2. Game Overview & Design Intent

Spit Shine is designed to occupy the space between pure-chance games (baccarat) and skill-heavy games (poker): simple enough for a casual player to learn in one round, deep enough for regulars to develop a real strategy. Every round contains exactly seven decision phases, and every phase creates observable information - shown cards, spit targets, shine behavior - that skilled players can exploit.

Design lineage

Feature	Poker	Blackjack	Baccarat	Spit Shine
Decks	1	6-8	8	2
Players vs	Each other	Dealer	Banker	All seats
Hand construction	5 cards	2+ cards	2-3 cards	4 cards to 2
Decision points	4	1	0	7 phases
Player information	Partial	1 up card	None	Partial + Spit intel
Bluffing	Core	None	None	Show/hide + Bow Out
Card exchange	Draw poker	None	None	Spit phase
Discard / redraw	None	None	None	Shine phase
Safety valve	Fold	Surrender	None	Bow Out (25% refund)
House edge source	Rake	Rules	Commission	25% pot rake + dealer Shine

Unique innovations

- Spit: exchange one face-down card with an opponent. You see what you receive; they do not see what you gave. Each seat can be spat at once per round.
- Shine: optional redraw. Players discard 0-1 face-down cards; the dealer may discard 0-2 - the house's structural edge.
- Bow Out: hidden hands may fold and recover 25% of the ante, a risk-management lever that rewards discipline.
- Showing: revealing cards is a commitment (it locks out Bow Out) and an information weapon.

3. Game Modes & Player Configurations

Spit Shine supports two officially specified modes. Both use the same five-seat table geometry and the same seven-phase round, but the payout structures and the dealer's role differ. This section is the authoritative reference for how many players a table can seat, how empty seats are handled, and what each mode pays.

3.1 Competitive mode (default)

Competitive mode is a five-seat game: four player seats and the dealer, all of whom ante 100 chips. The dealer is a full participant - the dealer can win the pot like any seat - but is barred from Bow Out and Spit, and may discard two cards in Shine instead of one. The highest-scoring hand at showdown wins 75% of the pot after the 25% rake.

Item	Specification
Seats	4 players + dealer (5 seats)
Ante	100 chips per seat, including the dealer
Pot (full table)	500 chips
Winner receives	375 chips (pot - 25% rake of 125)
Dealer restrictions	No Bow Out, no Spit; may Shine 0-2 cards
Example	YOU wins with K+Q (350). Pot 500, rake 125, payout 375. YOU nets +275 chips for the round.

3.2 Cooperative mode (players vs dealer)

Cooperative mode seats four players against the dealer. The dealer antes 0, so the pot is 400 chips. If any player beats the dealer at showdown, all non-bowed players share the 300-chip remainder after the 100-chip rake. If the dealer wins, the house keeps the entire pot. This mode's current economics are severely house-favorable and must be rebalanced before a casino deployment (see Section 13.5).

Item	Specification
Seats	4 players + dealer (5 seats)
Ante	100 chips per player; dealer antes 0
Pot (full table)	400 chips
Team win	All non-bowed players split 300 chips (75 each)
Dealer win	House keeps the full 400 chips
Audited team win rate	70.3% (20,000 rounds, seed 20260805)
Audited player EV	-46.1 chips per round (house edge ~46%)

Worked example: in a 10-round cooperative session, the team wins 7 rounds and the dealer wins 3. Each player's cash flow is $7 \times (-25)$ chips (pay 100, receive 75) plus $3 \times (-100) = -475$ chips, or -47.5 per round - consistent with the audited -46.1 before Bow Out refunds.

3.3 Player counts and seat configurations

Configuration	Seats	Use case	Notes
Digital POC / demo	1 human + 3 AI + dealer	Sales demo, training, simulation	AI fills seats; identical rules to live play
Live standard table	4 players + dealer	Recommended casino configuration	Full 500-chip pot in competitive mode
Live short-handed	2-3 players + dealer	Low-traffic hours	Empty seats do not ante; pot scales; requires approved rule amendment and published minimums
Minimum playable	1 player vs dealer	Cooperative mode only	Single-player cooperative is viable; competitive requires AI fill or a second player

Player-count limitations are operational, not mathematical: every seat plays an independent hand, so the rules and odds per seat are identical at any table size. A short-handed table simply has a smaller pot. Casinos should set a minimum of two players for competitive play and one for cooperative, publish the minimums, and keep empty seats clearly marked.

3.4 Mode selection rules

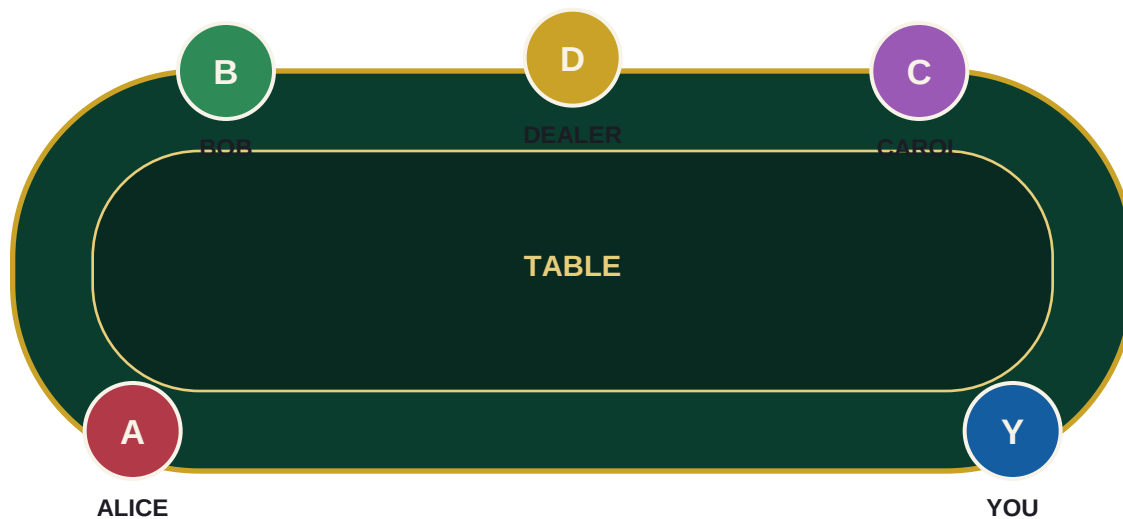
- Mode is announced before the first ante of a session and cannot change mid-shoe.
- A table may run different modes on different shifts, but a single table cannot mix modes in one round.
- Side wagers are not part of the current rules; any future side bet requires independent math certification before deployment.

4. Equipment & Table Layout

Equipment checklist

Item	Specification	Control requirement
Cards	Two unopened 52-card decks, identical backs	Inspect for marks; count 52 per deck before and after play
Shoe	Dealing shoe for up to 104 cards	Tamper-evident; fixed position under surveillance
Cut card	Standard plastic divider	Insert 13-15 cards from the bottom (approx. 85-87% penetration)
Chips	Standard 25/100/500 denominations	House drop box for rake; chip fills recorded
Table	5 seats + dealer, 8-12 ft oval	Dealer station with shoe, drop box, discard rack
Surveillance	Overhead + dealer camera	Cards, shoe, and drop box visible at all times

Seating and deal order



Deal order (canonical): YOU, ALICE, BOB, CAROL, DEALER - one card at a time

The canonical engine deals round-robin one card at a time in seat order: YOU, ALICE, BOB, CAROL, DEALER. A floor operation may adopt standard clockwise dealing starting left of the dealer and dealing the dealer last; the two orders are statistically equivalent. The Spit turn order follows the same fixed rotation in the reference implementation and should be rotated per round on a live floor (Section 13.7).

5. Card Values & Scoring

Rank point values

Rank	Points	Rank	Points	Rank	Points
2	20	7	70	Q	150
3	30	8	80	K	200
4	40	9	90	A	100 or 0 (duality)
5	50	10	100	A+A	1000 (Double Aces)
6	60	J	100		

Ace duality

An Ace counts as 100 points or as 0 points, whichever maximizes the final hand score. $A+K = 300$; $A+2 = 120$; A alone paired with nothing is never forced to count 0 unless it is the better option. Two Aces together are the special Double Aces hand: 1000 points, the highest possible score. The maximum non-Ace pair is $K+K = 400$, which means no non-Ace hand can reach the 500+ range.

Scoring examples

Hand	Calculation	Score
K + Q	$200 + 150$	350
K + A	$200 + 100$ (Ace = 100)	300
10 + J	$100 + 100$	200
A + 2	$100 + 20$ (Ace = 100 beats 0 + 20)	120
7 + 5	$70 + 50$	120
A + A	Special rule	1000

Ace duality in practice - further worked examples

Hand	Pairings considered	Best score	Why
A K 7 2	$A+K=300$, $A+7=170$, $A+2=120$, $K+7=270$, $K+2=220$, $7+2=90$	300	Ace counts 100 with the King
A 2 3 4	$A+4=140$, $A+3=130$, $A+2=120$, $4+3=70$, $4+2=60$, $3+2=50$	140	Ace as 100 dominates every non-Ace pair
A 9 8 2	$A+9=190$, $A+8=180$, $A+2=120$, $9+8=170$, $9+2=110$, $8+2=100$	190	Ace + 9 beats 9 + 8 by 20 points
K K 5 3	$K+K=400$, $K+5=250$, $K+3=230$, $5+3=80$	400	The maximum non-Ace score
Q Q 10 J	$Q+Q=300$, $Q+J=250$, $Q+10=250$, $10+J=200$	300	Pair of Queens beats 10+J by 100

Hand	Pairings considered	Best score	Why
A A 2 3	A+A=1000 (special), A+2=120, A+3=130, 2+3=50	1000	Double Aces overrides all other pairings

Exact hand-score distribution (best 2 of 4 cards)

Tier	Exact count (of 4,598,126)	Probability	Note
Monster - Double Aces (1000)	133,126	2.90%	
500-999	0	0.00%	Impossible by rule - no non-Ace pair reaches 500
Strong (350-499)	395,702	8.61%	
Solid (250-349)	1,480,934	32.21%	
Marginal (180-249)	1,583,762	34.44%	
Weak (100-179)	972,856	21.16%	
Dead (< 100)	31,746	0.69%	

Source: exact enumeration of all $C(104,4) = 4,598,126$ four-card hands in the canonical engine. Suits affect only tie-breaks, so this distribution is exact for scores.

6. Hand Evaluation & Tie-Breaks

Evaluation algorithm

- Step 1: If the hand contains two Aces, the score is 1000 (Double Aces).
- Step 2: For every other hand, score each Ace as 100 or 0 to maximize the total.
- Step 3: The score is the sum of the two cards' points under that rule.
- Step 4: From four cards, evaluate all six possible pairings and keep the best.

Showdown order (tie-breaks)

When two or more seats finish with the same score, the winner is resolved in this exact order: (1) higher score; (2) higher individual card rank; (3) higher suit value (clubs 1, diamonds 2, hearts 3, spades 4). If all three are identical, the pot is split equally among the tied seats.

Exact ties are not rare: 633 of 20,000 simulated rounds (3.16%) ended in a pot split. Any table procedure must define split payouts in advance (see Section 9.4).

Worked tie-break examples

Scenario	Hands	Resolution
Score tie, rank breaks it	ALICE: A+9 = 190 vs CAROL: 10+9 = 190	ALICE wins - high rank Ace (14) beats 10 (11)
Score and rank tie, suit breaks it	YOU: K(spades)+Q(clubs) = 350 vs ALICE: K(hearts)+Q(diamonds) = 350	YOU wins - highest suit value spades (4) beats hearts (3)
Full tie - pot splits	YOU: K(spades)+Q(hearts) = 350 vs ALICE: K(hearts)+Q(spades) = 350	Split - both score 350, rank 13, suit-max 4. Payout 375 becomes 187 + 187 + 1 chip rounding rake to the house

Audit note: earlier prototypes silently awarded the entire pot to the earliest seat on an exact tie. The canonical rule set adopted for casino operation splits the pot, matching the design documents.

7. How to Deal - Table Procedures

7.1 Pre-game preparation

- Open two new decks in front of surveillance; inspect every card for marks, bends, or irregularities; count 52 cards per deck.
- Shuffle: minimum seven riffle shuffles with alternating cuts, followed by a strip and a final riffle. Square the deck completely.
- Insert the cut card 13-15 cards from the bottom (recommended penetration 85-87% for a two-deck game).
- Load the shoe face-down, verify the deck's edge is even, and place the shoe in the marked position.
- Burn procedure: before the first round of a new shoe, draw one card face-down to the discard rack; it plays no part in the round.
- Announce table limits and confirm each seat's ante before dealing.

7.2 Round flow - dealer script

Step	Action	Dealer control
1	Collect antes (100 per active seat; dealer antes 100 in competitive mode, 0 in cooperative mode).	Place chips in pot area; verify count.
2	Deal four cards to each seat, one card at a time, in seat order; dealer takes the last card of each pass.	No card may be exposed; announce 'Cards out' when complete.
3	Phase 1: allow private review. No table action.	Do not touch cards.
4	Phase 2: seats may show 0-4 cards. Announce each shown card as it is revealed; remind players that showing locks out Bow Out.	Record shown cards on the audit pad.
5	Phase 3: hidden seats may Bow Out. Pay 25% of ante (25 chips) and muck their cards.	Remove the refund from the pot; verify pot total.
6	Phase 4: each active non-dealer seat in turn may Spit once. Execute the swap without exposing the given card; reveal the received card to the spitter only.	Mark the target seat as 'spat' - one Spit per target per round.
7	Phase 5: seats may Shine (discard 0-1 face-down and draw). The dealer Shines last with 0-2 discards.	Draw replacements from the shoe; discard the old cards.
8	Phase 6: each seat selects its best two-card hand; the dealer selects their own pair.	Confirm no seat has more or fewer than four cards.
9	Phase 7: dealer reveals their own pair first, then each seat's pair; announce scores and determine the winner using score, rank, then suit.	On an exact tie, split the pot per Section 9.4.
10	Payout: rake 25% of the pot to the drop box; pay the winner(s); collect and count the discard rack.	When the cut card appears, finish the round and reshuffle.

7.3 Shoe and security controls

- The cut card ends the shoe: stop dealing after the round in which it appears, then reshuffle using the same procedure.
- The discard rack must be visible to surveillance at all times; discards are never returned to the shoe mid-shoe.
- A dealer change or table opening requires a full shuffle and a new burn card.

- Any exposed card during the deal is treated as a dead card: remove it to the discard rack and redeal that seat's card from the shoe.

7.4 Worked dealing examples

Example	Detail	Result
Shoe penetration	Cut card inserted 14 cards from the bottom of a 104-card shoe.	90 playable cards = 86.5% penetration. At ~22 cards per round (20 dealt + ~2 Shine draws), the shoe supports about four rounds before the cut card appears.
Shuffle sequence	Seven riffle shuffles with alternating cuts, a strip, and a final riffle, timed at roughly two minutes for a two-deck shoe.	Meets the minimum mixing requirement; the sequence is logged with the round audit for regulator review.
Dead card during deal	On the third pass, the dealer exposes the 5 of hearts while dealing to BOB.	The 5 of hearts is placed in the discard rack as a dead card and BOB receives the next card from the shoe. The round continues; no redeal.
Burn card	First round of a new shoe: one card is burned face-down to the discard rack before antes are collected.	The burned card plays no part in any round of that shoe.

8. How to Play - Phase-by-Phase Player Guide

Each round follows seven phases. The player guide below describes what happens, what actions are legal, and what the dealer does in each phase, followed by the strategy note used by the in-game coach.

Phase 1 - Private Reveal

Aspect	Detail
What happens	Four cards are dealt face-down to each seat. Only you can see your own cards.
Your options	No action required. Review your hand and form a plan.
Dealer's role	Dealer has no role beyond the deal.
Coach tip	Assess your best two-card pairing and classify it: Strong (350+), Solid (250-349), Marginal (180-249), Weak (below 180).

Phase 2 - Showing

Aspect	Detail
What happens	You may turn 0-4 of your cards face-up on the table. Shown cards are visible to every seat for the rest of the round.
Your options	Toggle any combination of face-down cards face-up, then confirm. Showing ANY card permanently locks out Bow Out.
Dealer's role	Dealer announces and records shown cards.
Coach tip	Strong hands often show one card to deter Spits; weak hands hide everything to keep Bow Out available. Showing is a commitment - never show a hand you are not willing to play.

Phase 3 - Bow Out

Aspect	Detail
What happens	If you showed no cards, you may forfeit the round and recover 25% of your ante (25 chips on a 100-chip ante).
Your options	Bow Out or stay. Bowing seats are removed from the round; their remaining pot contribution stays.
Dealer's role	Dealer processes the refund from the pot and mucks the cards.
Coach tip	Break-even math: staying costs 100, Bowing costs 75. Stay only if your win probability exceeds about 6.7%. Hands below roughly 150 points should normally Bow Out.

Phase 4 - Spit

Aspect	Detail
What happens	Each active non-dealer seat receives one Spit: give one face-down card to a target and take one of the target's face-down cards.

Aspect	Detail
Your options	Choose your card to give and your target. Each target can be spat at once per round. You may also pass.
Dealer's role	Dealer executes the swap, reveals the received card to the spitter only, and marks the target as spat.
Coach tip	Spit dumps your weakest card and draws a random replacement (expected value about 81 points). Simulations show Spitting is worth roughly 9 chips per round versus never Spitting. Strong hands (350+) usually pass to avoid disruption.

Phase 5 - Shine

Aspect	Detail
What happens	You may discard 0-1 face-down cards and draw replacements from the shoe. The dealer may discard 0-2.
Your options	Choose a face-down card to discard, or keep your hand. You draw only if you discard.
Dealer's role	Dealer discards and draws for each seat, then for the dealer seat.
Coach tip	Below 180 points, Shine is usually correct. Above 250, keep. The dealer's extra discard is the house's structural edge - plan around it.

Phase 6 - Selection

Aspect	Detail
What happens	From your four cards, select the two-card pairing you will play at showdown.
Your options	Choose any two cards. The selection is final.
Dealer's role	Dealer selects the dealer pair.
Coach tip	Always consider all six pairings. The engine recommends the maximum score; remember Aces count 100 or 0 to maximize.

Phase 7 - Showdown & Payout

Aspect	Detail
What happens	All remaining seats reveal their pairs. The highest score wins; ties split.
Your options	No player action - results are announced.
Dealer's role	Dealer reveals first, then each seat; announces scores, applies tie-breaks, rakes 25%, and pays the winner(s).
Coach tip	Watch the dealer's revealed pair every round - it is the single most informative datum at the table.

9. Dealer Procedures & Payout Math

9.1 Pot accounting

The pot is the sum of all antes minus Bow Out refunds. At showdown the house rakes 25% (floor of the integer division) and the remainder is paid to the winner, or split among tied winners. Any chips left over after an equal split (for example, splitting 375 two ways leaves 1 chip) are retained by the house as rounding rake - this rule must be posted at the table.

9.2 Payout examples

Scenario	Pot	House rake (25%)	Winner(s)	Payout
Full table, no Bow Outs	500	125	One winner	375
One Bow Out (25 refund)	475	118	One winner	357
Two Bow Outs (50 refunds)	450	112	One winner	338
Exact tie, full table	500	125	Two winners	187 each + 1 rounding rake
Cooperative, players win	400	100	Four players	75 each (300 total)
Cooperative, dealer wins	400	-	Dealer	House keeps entire pot (400)

Rounding rake note: $500 \times 25\% = 125$, leaving 375. Splitting 375 between two winners yields $187 + 187 + 1$ remainder, which is retained by the house and logged. All payouts must be announced before chips move.

9.3 Dealer's own play

The dealer is a full seat in competitive mode (antes 100, can win the pot) but never Bows Out and never Spits. The dealer's structural advantage is the two-card Shine: in 20,000 simulated rounds the extra discard is worth roughly 6.8 points of win rate, lifting the dealer to a 24.6% win rate versus a fair 20%. Dealers follow the same thresholds as the reference AI: Shine 2 cards below 200 points, 1 card at 200-299, keep at 300+.

9.4 Disputes and corrections

- Any exposed card during the deal is a dead card: discard it and draw a replacement for that seat.
- If a player exposes a face-down card during Spit or Shine, the card is still played (it becomes known information); the floor decides penalties per house policy.
- Payout disputes are resolved from the audit log; every round's deal, show, bow, spit, shine, selection, and payout events are recorded with card-level detail.

9.5 Worked payouts with chip values

Chip value	Ante	Pot after one Bow Out	Rake (25%)	Winner payout
\$0.25	\$25	475 chips = \$118.75	118 chips = \$29.50	357 chips = \$89.25
\$1.00	\$100	475 chips = \$475.00	118 chips = \$118.00	357 chips = \$357.00
\$5.00	\$500	475 chips = \$2,375	118 chips = \$590	357 chips = \$1,785

Worked split example at \$1 chips: full table, two winners split 375. Each winner receives 187 chips (\$187); the remaining 1 chip (\$1) goes to the house as rounding rake and is recorded in the audit log.

10. Limits, Bankroll & Money Management

This section defines table limits, buy-in and bankroll guidance, and the cash and chip controls a floor operation needs. Limits are expressed in chips first, then in currency, because the chip value is the operator's calibration lever.

10.1 Recommended table limits by chip value

Chip value	Standard ante	Min buy-in	Max ante	Max single-round win
\$0.25	100 chips = \$25	20x ante = \$500	500 chips = \$125	375 chips = \$93.75
\$1.00	100 chips = \$100	20x ante = \$2,000	500 chips = \$500	375 chips = \$375
\$5.00	100 chips = \$500	20x ante = \$10,000	2,000 chips = \$10,000	375 chips = \$1,875

The maximum single-round win is the full pot minus rake: at a full table with a \$100 ante, that is 375 chips. A player can never win more than the pot, so no separate jackpot exposure exists unless a future side wager is added (which would require its own certification).

10.2 Player bankroll guidance

Guideline	Rule of thumb	Worked example (\$1 chips)
Session buy-in	At least 20x the ante	\$100 ante -> \$2,000 buy-in
Session loss limit	Stop at 5x the ante	Stop-loss at \$500 per session
Bow Out discipline	Fold below ~150 points when hidden	Saves 25 chips per fold; at ~9% of rounds this adds ~2.3 chips per round versus never folding
Expected round cost	Audited EV is -33.9 chips per round at seat 0 (competitive)	A \$2,000 bankroll supports roughly 50-60 rounds on average; variance dominates short sessions

10.3 Cash and chip flow

Funds move in a closed loop: player buy-ins exchange cash for chips from the tray; antes move chips to the pot; at showdown the rake moves to the drop box and the payout moves back to the winner's stack; Bow Out refunds return to the player. At end of shift the chip equation must balance.

Reconciliation item	Amount	Worked example
Opening float (tray)	+\$2,000	Chips issued to the table
Player buy-ins (cash)	+\$4,000	Cash exchanged for chips
Closing tray	-\$2,100	Chips still in the tray
Drop box (cash removed)	-\$3,900	Cash moved to the cage
Check	$\$2,000 + \$4,000 = \$2,100 + \$3,900$	Balanced: $\$6,000 = \$6,000$
Audit theoretical win	\$1,100	Sum of per-round rakes recorded by the audit log
Variance monitoring	+2.9%	Actual win \$1,132 vs theoretical \$1,100 - within the standard +/-3% tolerance; investigated only if outside

10.4 Responsible gaming and jurisdictional limits

- Publish table minimums and maximums at the table and in the house rules; they are configurable per jurisdiction and per venue.
- Support player-set loss limits, time limits, and voluntary exclusion through the standard responsible-gaming tools.
- Currency transaction reporting and AML thresholds apply at the cage, not the table; the dealer never handles cash directly - all buy-ins are exchanged through the tray under surveillance.

11. Daily Turnover & Revenue Model

This section models the funds that flow through a Spit Shine table and the revenue it can generate. All figures are illustrative and depend on four operator-set variables: chip value, round duration, table occupancy, and operating hours. The audit supplies the fixed part: the house keeps 24.5% of the pot (rake) in competitive mode.

11.1 Definitions

Term	Definition	Spit Shine value
Handle (gross turnover)	Total antes placed across all seats in a period	500 chips per full competitive round
Table win (hold)	House revenue before expenses	Rake only: 24.5% of pot in competitive mode
Hold percentage	Table win / handle	~24.5% competitive; ~46.1% cooperative (flagged)
Drop	Cash and chips exchanged at the table	Tracked per shift; reconciled against theoretical win
Rounds per hour	Dealt rounds per hour of operation	Model range 17-30 (see assumptions)

11.2 Model assumptions (must be validated in floor trials)

Variable	Conservative	Standard	Aggressive
Round duration	3.5 minutes	2.5 minutes	2.0 minutes
Rounds per hour	17	24	30
Operating hours / day	12	12	24
Average occupancy	75% (3 of 4 player seats)	75%	75%
Chip value	\$1	\$1	\$1
House win per full round	\$122.50 (audited)	\$122.50	\$122.50

11.3 Per-round economics

Configuration	Handle per round	Expected house win	Winner payout
Competitive, full table (4 players + dealer)	500 chips = \$500	\$122.50 (24.5%)	375 chips = \$375
Competitive, 3 players + dealer	400 chips = \$400	\$98.00 (24.5%)	300 chips = \$300
Cooperative, full table (4 players, dealer antes 0)	400 chips = \$400	\$184.40 (46.1% - flagged)	75 chips per winning player

11.4 Daily turnover scenarios (worked)

Scenario	Rounds per day	Handle	House win (24.5%)
Conservative: 17 rph x 12 h	204	204 x \$500 x 0.75 = \$76,500	\$76,500 x 24.5% = \$18,743
Standard: 24 rph x 12 h	288	288 x \$500 x 0.75 = \$108,000	\$108,000 x 24.5% = \$26,460

Scenario	Rounds per day	Handle	House win (24.5%)
Aggressive: 30 rph x 24 h	720	$720 \times \$500 \times 0.75 = \$270,000$	$\$270,000 \times 24.5\% = \$66,150$

The occupancy factor of 0.75 models an average of three of four player seats in use (handle scales with occupied seats). At full occupancy, multiply by 1.33; at 50% occupancy, multiply by 0.67. All figures scale linearly with chip value: at \$0.25 chips, divide by 4; at \$5 chips, multiply by 5.

11.5 Monthly and multi-table projections

Deployment	Handle / day	House win / day	House win / 30 days
1 table, standard	\$108,000	\$26,460	\$793,800
5 tables, standard	\$540,000	\$132,300	\$3,969,000
10 tables, standard	\$1,080,000	\$264,600	\$7,938,000

These projections assume the model's standard case and full-year operation at the modelled occupancy. They are deliberately striking because a 25% rake is far above mainstream table games (blackjack hold is typically 10-25% of drop but with much lower edge per bet; baccarat holds 10-15%). The rake is the game's key commercial lever: Section 11.7 shows calibration options.

11.6 Player session example (funds flow)

Round	Result	Player cash flow	Running total
1	Win (full table)	+257	+257
2	Loss	-100	+157
3	Bow Out	-75	+82
4	Loss	-100	-18
5	Win (full table)	+257	+239
6	Loss	-100	+139
7	Bow Out	-75	+64
8	Loss	-100	-36
9	Loss	-100	-136
10	Loss	-100	-236

Session summary: \$1,000 buy-in, 10 rounds, 2 wins, 6 losses, 2 Bow Outs. Net = $2 \times (+257) + 6 \times (-100) + 2 \times (-75) = +514 - 600 - 150 = -236$ chips (-\$236 at \$1 chips). This is an illustrative sequence, not a prediction.

11.7 Calibration options for market fit

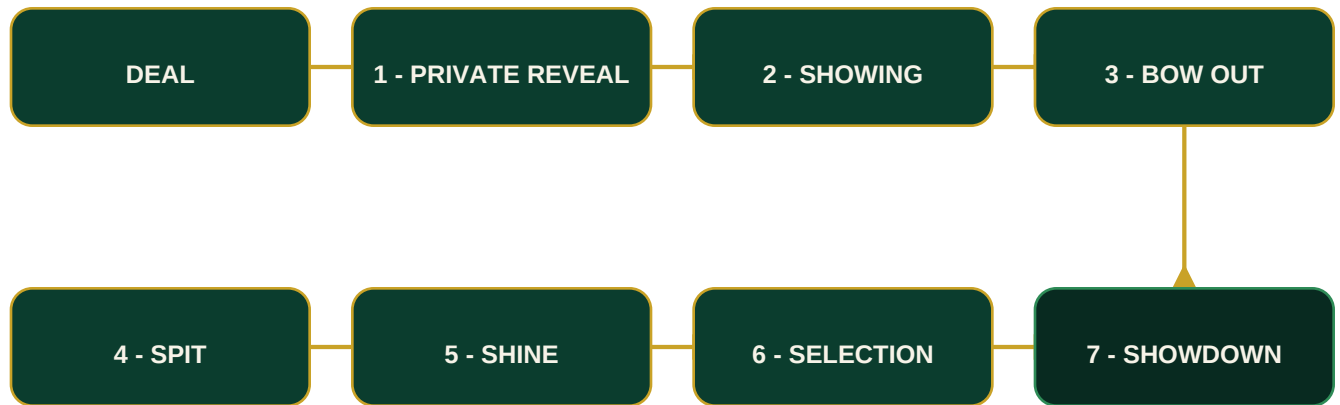
Rake	Hold (approx.)	House win per full round	Note
25% (current)	24.5%	\$122.50	Audited baseline; very high for a table game
15%	14.7%	\$73.50	Closer to premium table-game norms
10%	9.8%	\$49.00	Competitive with popular carnival games

The operator chooses the rake and chip value as a package. A lower rake improves player retention and table longevity; a higher chip value restores revenue per table. Both variables must be settled before the independent

game-math certification, because they change the certified payout tables.

12. Game State Machine & Canonical Rules

Round state flow



Canonical rules reference (v11)

Rule	Specification
Shoe	104 cards (2 decks), fresh and shuffled every round
Deal	4 cards per seat, round-robin, one card at a time
Ante	100 per seat; dealer antes 0 in cooperative mode
Showing	0-4 cards; showing any card locks out Bow Out
Bow Out	Hidden seats only; refund 25% of ante; dealer never bows
Spit	One per non-dealer seat; each target once per round; received card private to spitter
Shine	0-1 discards (players), 0-2 (dealer); draw only after discard
Selection	Best 2 of 4; human may choose any two cards
Showdown	Score, then high rank, then high suit; exact ties split
Payout	Rake 25% of pot; winners share remainder; rounding rake to house
Modes	Competitive (all seats) and Cooperative (players vs dealer)

State invariants

- A seat can Bow Out only once and only while hidden.
- A seat can Spit at most once; a target can receive at most one Spit.
- Spit and Shine operate only on face-down cards.
- The pot never goes negative; refunds are taken from the pot before rake.
- The dealer is always active at showdown (never Bows Out).

- Every action is written to the audit log before the next action begins.

13. Audit Data - Odds, EV & Fairness

All figures below are produced by the canonical engine v11 and the audit harness (spitshine_analysis.py). Exact probabilities are enumerated combinatorially; round-level figures come from 20,000 Monte Carlo rounds per scenario with a fixed seed (20260804), playing the documented threshold policies at every seat. Chip conservation (sum of seat nets + house take = 0) holds to machine precision.

13.1 Exact probabilities

Event	Probability	Method
A specific 2-card pair is A+A	0.52%	Exact - 28 of $C(104,2) = 5,356$ pairs
Your 4-card hand contains A+A	2.90%	Exact hypergeometric
At least one seat dealt A+A in a round	13.92%	Monte Carlo - 200,000 deals
At least one seat finishes with A+A after Spit/Shine	17.2%	20,000-round simulation
Best pair 350+ (any seat's deal)	11.50%	Exact hand enumeration

The Spit and Shine phases create roughly 3.3 points of additional Double-Aces frequency because a lone Ace can be paired by a redraw or exchange. This is a deliberate feature of the game's variance profile.

13.2 Round outcomes - competitive mode (20,000 rounds)

Seat	Win rate	Avg final score	Net EV per round
YOU (seat 0)	17.8%	241.5	-33.9
ALICE (seat 1)	19.3%	246.7	-28.7
BOB (seat 2)	20.0%	249.0	-26.5
CAROL (seat 3)	21.4%	255.3	-21.0
DEALER (seat 4)	24.6%	278.0	-12.5
House (rake)	-	-	+122.5 per round

Win rate by seat - competitive (fair baseline 20%)



Every seat - including the dealer - has negative expected value in competitive mode; only the house rake is positive. The dealer's 24.6% win rate is the strongest structural advantage in the game, and it comes almost entirely from the two-card Shine.

13.3 Winner score distribution

Winning score	Rounds	Share
Double Aces (1000)	3,442	17.2%
350-499	7,307	36.5%
250-349	8,777	43.9%
180-249	474	2.4%

13.4 House edge - corrected mathematics

With a full 500-chip pot and a fair 20% win rate, the winner nets $375 - 100 \text{ ante} = +275$ chips. Expected value is therefore $E = 0.20 \times 275 - 0.80 \times 100 = -25$ chips per round. That is a 25% house edge per ante, exactly the 25% rake - not the 5% claimed in the original white paper, which incorrectly treated the gross 375 payout as profit. Measured house take in simulation is 122.5 chips per round (24.5%) after Bow Out refunds.

Break-even question	Correct answer	Formula
Stay in vs Bow Out	~6.7% win chance	$25 / 375$
Break even for the round	~26.7% win chance	$100 / 375$
Fair-game EV (20% win rate)	-25 chips per round	$0.2 \times 275 - 0.8 \times 100$

Worked EV calculations

Perspective	Formula	Result	Measured (20k rounds)
Player, seat 0	$0.1783 \times 275 - 0.8217 \times 100$	-33.1 chips / round	-33.9 chips / round
Dealer seat	$0.2461 \times 275 - 0.7539 \times 100$	-7.7 chips / round	-12.5 chips / round

Perspective	Formula	Result	Measured (20k rounds)
House	Rake only (25% of pot)	+125.0 chips / round (full pot)	+122.5 chips / round
Bow decision, 12% win chance	$0.12 \times 275 - 0.88 \times 100$	-55.0 -> stay (bow costs -75)	-
Bow decision, 5% win chance	$0.05 \times 275 - 0.95 \times 100$	-81.3 -> bow (saves 6.3)	-

The small gap between formula and measured EV comes from pot-size variation: Bow Out refunds shrink the pot, so the actual winner payout is below the 375-chip full-pot figure used in the closed-form formula.

13.5 Cooperative mode economics and rebalancing

In cooperative mode the dealer antes 0, the pot is 400, and the team of four players wins if any player beats the dealer. The team wins 70.3% of rounds in simulation, yet each player still nets -46.1 per round because a team win pays only 75 chips gross per player, and a dealer win forfeits the entire pot. The house takes 184.4 chips per 400-pot (46.1% effective edge). This mode is not viable for a casino floor in its current form and is flagged for rebalancing.

Rebalance package (illustrative, pending certification)

Option	Change	Audited/derived effect
A - Reduce dealer Shine in cooperative	Dealer discards 0-1 cards like players	Team win rate rises from 70.3% to 78.6% (20,000-round experiment, seed 20260806)
B - Dealer antes + higher team payout	Dealer antes 100; team win pays 125 gross per player (pot 500)	Player EV improves from -46.1 to about -12.1 chips/round
C - Combined package	Both A and B	Player EV about -1.8 chips/round - near fair, comparable to standard house-banked games

Option C is the recommended starting point for a casino deployment: it removes the dealer's double-discard edge in this mode and makes the payout structure fair while keeping the house positive. It must be formally simulated and certified before publication.

13.6 Rule variants - what each mechanic contributes

Variant	Dealer win	YOU win	YOU net/round
Baseline (canonical)	24.6%	17.8%	-33.9
Dealer shines 1 card (like players)	18.3%	18.9%	-30.1
Dealer also Spits (legacy v4 rule)	29.7%	17.1%	-36.1
Spit order rotates each round	24.9%	18.9%	-30.3
Human never Spits	26.1%	15.3%	-42.9
Nobody Shines	17.8%	18.8%	-30.6
Conservative 'smart' human	25.1%	15.3%	-36.3

Reading the variants: the dealer's second Shine discard is worth about 6.8 points of win rate (24.6% to 18.3%); the legacy v4 rule that let the dealer Spit pushed the dealer to 29.7%; rotating the Spit order equalizes the player seats; and never Spitting costs the player roughly 9 chips per round.

13.7 Seat-position fairness

With a fixed Spit order, seat 0 (YOU) wins 17.8% while seat 3 (CAROL) wins 21.4% - a systematic positional bias of about 3.7 points. Reversing the order inverts the bias, confirming the Spit sequence itself is the cause. The recommended floor fix is to rotate the Spit starting seat each round (poker-button style), which equalizes the four player seats without changing house economics. This must be included in the table procedures.

14. Integrity, Fairness & Regulatory Readiness

11.1 Randomness

The reference implementation uses a seeded Mersenne Twister for reproducibility and testing. A live casino deployment requires a certified hardware RNG (NIST SP 800-90A, FIPS 140-2/3 validated) with continuous statistical testing (NIST STS, Dieharder) and documented seeding for every shuffle.

11.2 Audit trail

Every round is recorded with card-level detail: deal, showing, Bow Out refunds, Spit swaps (who gave what to whom), Shine discards and draws, pair selections, showdown scores, tie-breaks, rake, and payouts. The log is exported as JSON and CSV from the client (AUDIT button) and is the basis for dispute resolution and regulator reporting.

11.3 Regulatory readiness checklist

Requirement	Status	Owner
Single canonical rules engine	In place (v11)	Development
Certified hardware RNG + daily statistical tests	Open	Vendor / IT
Card-level audit log with integrity (HMAC/keyed)	Log in place; keying open	Development
Shuffle / penetration / burn procedures	Documented in this guide	Operations
Independent game-math certification (e.g., GLI)	Open	Testing lab
Table limits, dispute and surveillance procedures	Open	Operations
Spit order rotation for positional fairness	Recommended - implement in floor rules	Operations
AML / KYC / responsible-gaming integration	Open	Compliance
Cooperative mode economics rebalanced or withdrawn	Open	Product

15. Known Issues & Remediation

The audit identified defects across the earlier prototypes. All gameplay-critical defects are fixed in the canonical engine v11; the register below is retained for transparency and for any laboratory reviewing the development history.

ID	Severity	Issue	Status
B1	Critical	v7 GUI human Spit targeted self; v7 sim Spit was a no-op	Fixed in v11
B2	Critical	v9/v10 editions imported an unavailable copperhead module and could not launch	Superseded by v12 client
B3	High	Spat-at-once rule tracked the wrong set - multiple Spits per target possible	Fixed in v11
B4	High	Fixed Spit order created seat-position bias	Rotation recommended
B5	High	AI Bow Out missing in interactive v7/v10	Fixed in v11
B6	High	White paper odds/EV errors (Double Aces, house edge, dealer EV)	Corrected in this guide
B7	Medium	Exact ties awarded to earliest seat instead of split	Fixed in v11
B8	Medium	Cooperative all-Bow-Out crash	Fixed in v11
B9	Medium	Cooperative house edge ~46% undisclosed	Flagged for rebalance
B10	Medium	Dealer Spit inconsistent between v4 and v7/v9/v10	Canonical: dealer never Spits
B11	Low	Tie-break suit semantics inconsistent (max vs sum)	Fixed in v11
B12	Low	Integer split remainder unaccounted	Defined as rounding rake

16. Appendix A - Glossary

Term	Definition
Ante	Fixed wager placed by each seat at the start of a round (default 100).
Bow Out	Phase 3 option to forfeit and recover 25% of the ante.
Dealer	Seat 4; never Bows Out, never Spits, may Shine 0-2 cards.
Double Aces	A+A pair worth 1000 points, the highest hand.
Face-down / Face-up	Hidden versus table-visible card state.
House rake	25% of the pot taken at showdown.
Pot	Sum of antes minus Bow Out refunds.
Shine	Phase 5 discard-and-redraw; players 0-1, dealer 0-2.
Showdown	Phase 7 comparison: score, high rank, high suit, split.
Spit	Phase 4 exchange: give one face-down card, take one; received card private to the spitter.
Shoe	The 104-card dealing container.
Split	Exact-tie resolution that divides the pot equally.

17. Appendix B - Worked Example: One Full Round

The round below is a fully scripted competitive round that exercises every mechanic: Showing, Bow Out, Spit, Shine, tie-breaking, rake, and payouts. Cards are fixed for illustration; in live play every card is dealt from the shoe.

B.1 Initial deal

Seat	Cards dealt	Best pair	Tier
YOU	K(spades) Q(diamonds) 7(clubs) 3(hearts)	K+Q = 350	Strong
ALICE	A(hearts) 9(spades) 8(diamonds) 2(clubs)	A+9 = 190	Marginal
BOB	J(hearts) 7(diamonds) 5(clubs) 4(spades)	J+7 = 170	Weak
CAROL	10(spades) 6(hearts) 5(diamonds) 3(clubs)	10+6 = 160	Weak
DEALER	K(clubs) 10(hearts) 8(diamonds) 3(spades)	K+10 = 300	Solid

B.2 Phase decisions

Phase	Decision	Why
2 - Showing	YOU shows K(spades); ALICE, BOB, CAROL hide; DEALER shows K(clubs)	YOU and the dealer have 350/300 hands and show to deter Spits; the marginal hands hide to keep Bow Out
3 - Bow Out	BOB Bows Out (refund 25); ALICE and CAROL stay	BOB's 170-point hand is below the fold line; ALICE and CAROL stay despite weak hands
5 - Shine	CAROL discards 2(clubs), draws 9(clubs); everyone else keeps	CAROL's hand is 180 after her Spit; discarding the 2 and drawing 9(clubs) lifts her best pair to 10+9 = 190. YOU/ALICE/dealer hold 350/190/300 and keep

B.3 Spit swaps (Phase 4)

Actor	Target	Card given	Card taken	Hand after swap
YOU	ALICE	3(hearts)	8(diamonds)	K Q 7 8 - best pair still 350
ALICE	CAROL	2(clubs)	5(diamonds)	A 9 3 5 - best pair still 190
CAROL	DEALER	3(clubs)	8(diamonds)	10 6 2 8 - best pair 180 (improved)

Each target receives exactly one Spit: ALICE from YOU, CAROL from ALICE, DEALER from CAROL. BOB is out of the round and cannot be targeted. The dealer never Spits.

B.4 Showdown and payouts

Rank	Seat	Pair	Score	Notes
1st	YOU	K + Q	350	Winner - pot 475, rake 118, payout 357, net +257

Rank	Seat	Pair	Score	Notes
2nd	DEALER	K + 10	300	Loses 100
3rd	ALICE	A + 9	190	Beats CAROL on high rank (Ace 14 vs 10)
4th	CAROL	10 + 9	190	Tie with ALICE on score, loses on rank
Out	BOB	-	-	Bow Out: net -75

Cash-flow check: +257 (YOU) - 100 (ALICE) - 100 (CAROL) - 100 (dealer seat) - 75 (BOB) + 118 (house rake) = 0. Every chip is accounted for, and the same check is performed automatically by the audit log after every round.

18. Appendix C - Data Provenance & Reproduction

Artifact	Path / command
Canonical engine	spitshine_engine.py
Audit harness	spitshine_analysis.py --rounds 20000
Raw audit data	spitshine_odds_report.json
Headless integration test	test_aaa_flow.py
Playable client + Academy	spitshine_aaa.py
Full audit report	SPIT_SHINE_AUDIT_REPORT.md (SS-AUDIT-2026-001)

All Monte Carlo figures use the fixed seed 20260804 (competitive baseline and variants) with 20,000 rounds per scenario, and the canonical threshold policies at every seat. Exact probabilities are computed by enumeration of all $C(104,4)$ hands. Chip conservation holds to machine precision in every run.

End of document. SS-GUIDE-2026-001 v1.0 - approved for distribution to casino business development and regulatory review.